Hands-on: Three Operator Cases

One notebook. Three operator problems.

90 minutes / three short exercises

Alaa Selim
MIT/Dartmouth

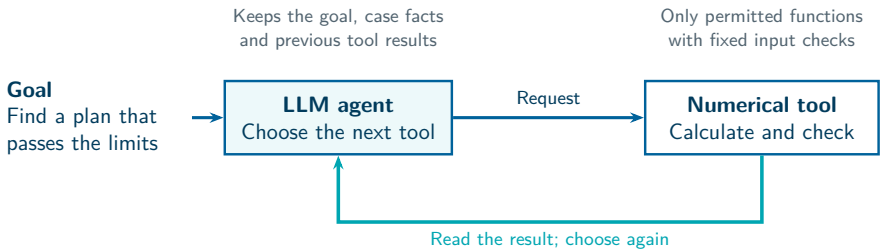
Mohamed Chehada
UT Austin

IEEE ComSoc IEEE Communications Society | IEEE

Independent educational material / IEEE ComSoc theme adaptation

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

What is agentic AI?



Teaching diagram. Llama choices are saved; Python calculations can run now.

- An AI agent chooses a tool, reads its result and chooses the next step toward a goal.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.2.

A tool call and an agent workflow

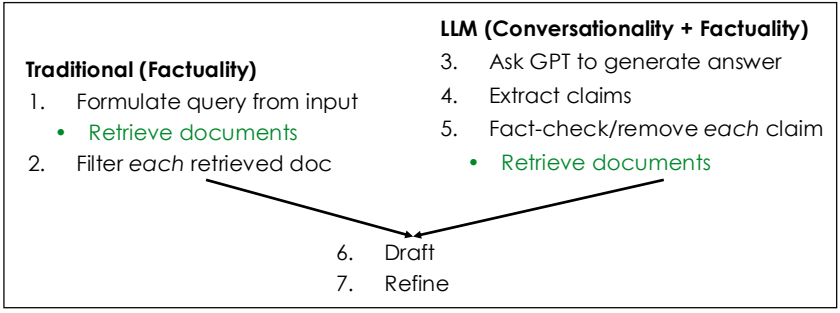


Source figure, p.2. Function calling and a multi-step workflow.

- A tool call does one job. An agent links jobs using the returned results.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.2.

Evidence before an answer



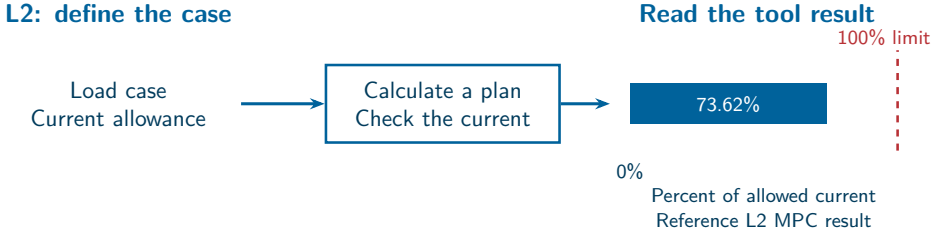
Source figure, p.19. WikiChat checks claims before writing.

- WikiChat checks claims before writing. Our lab checks numbers before reporting a result.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.19.

Voltage: propose, then check

Lab preview: L2



Teaching adaptation. L2 runs Python; L10B replays saved model choices.

- L2 calculates and checks proposals. L10B shows saved Llama tool choices.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.19; Classroom Labs L2.

Clear rules for tool use

- **Key Idea:** Full specifications are necessary
  - Not just functions to call!
- **Genie Worksheet: A Domain-Specific Language**
  - Specification of knowledge corpora + task workflow
    - Similar to workflow on webpages
  - Run-time to manage the long context

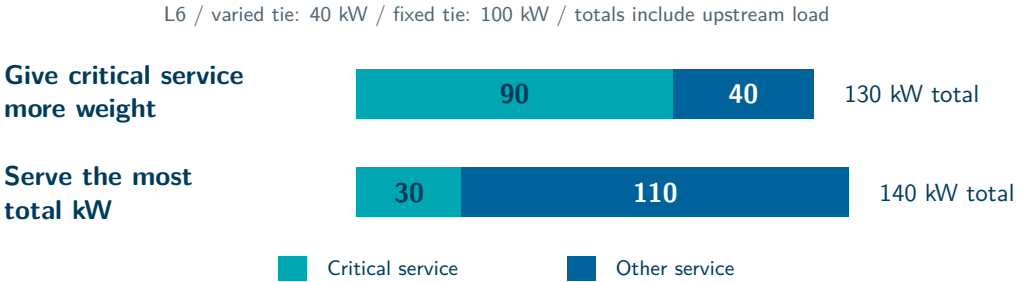
Source figure, p.40. Task rules define permitted tool use.

- Set the inputs, allowed tools, limits and stop rules.
- Python keeps these rules fixed when Llama chooses a step.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.40.

Restoration: define the service goal

Lab preview: L6



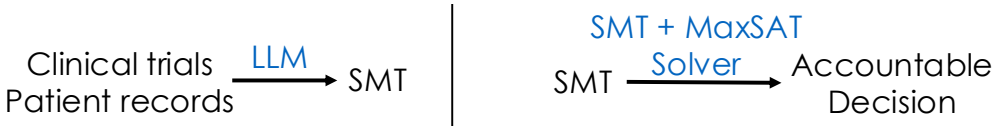
L6: same network and limits, different service goals.

- The goal changes which loads the solver restores. L6 compares two goals.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.40; Classroom Labs L6.

Language and decision checks

Source example / grid application



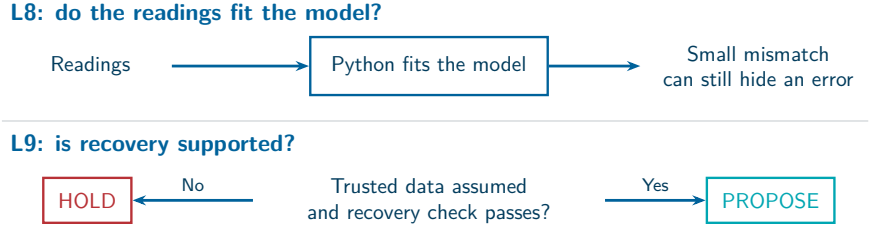
Source figure, p.53. Clinical example using logical constraints, not a power-system model.

- The source example turns written facts into constraints for a solver.
- In our lab, Llama requests a study. Python checks it. A person reviews it.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.53.

Cyber: good fit can hide bad data

Lab preview: L8–L9



L9 assumes those extra readings are trusted. The numerical tool does not authenticate them.

L8 checks fit. L9 requires a trust assumption before proposing recovery.

- L8 tests misleading readings. L9 shows when to return HOLD or a proposal.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.53; Classroom Labs L8–L9.

Four questions about any result

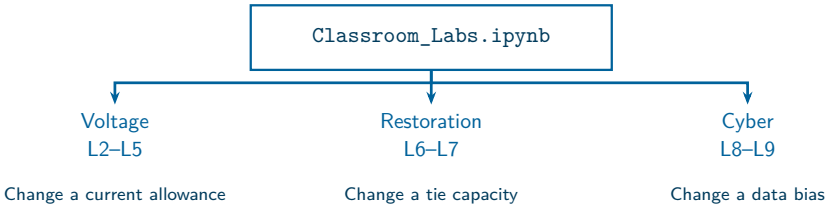
1. **Evidence**
   Which tool result supports the claim?
2. **Assumptions**
   Which inputs and model were used?
3. **Rules**
   Were the same limits and priorities kept?
4. **Next step**
   What does the operator still need to review?

Teaching adaptation of the cited lecture. The lab uses custom Python.

- Which tool ran? What did it assume? Which limits passed? What should the operator do next?

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, pp.52–53.

One notebook for all three cases



L10B replays Llama choices in the same file. Each case uses a separate teaching network.

Classroom Labs is the only required notebook.

Open Classroom Labs in Google Colab

- Open Classroom Labs and save a copy in Drive. Use it throughout.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Colab setup



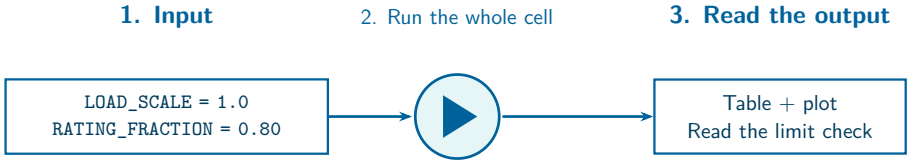
For the guided workshop, run each named lesson when prompted. Run all is a useful setup check.

Setup first, then L0 and L1.

- Save a copy, connect a CPU and run setup.
- Wait for Ready, then run L0 and L1.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Input, run, output



The play button executes Python. The saved text below a cell is its output.

Illustration of a code cell. Run the full L2 cell.

- Find the L heading and run its whole code cell.
- Read the output below it. After any edit, rerun the cell.

Source: Google Colab FAQ; Classroom Labs L2, cell classroom-007.

L0: keep the default settings

Classroom Labs / L0 / classroom-002

Classroom_Labs.ipynb L0 (classroom-002), source lines 11–14.

```
RUN_OPENDSS = False       # True: fresh full-model audit +
   separate AC MPC solve in L5
RUN_LOCAL_LLAMA = False   # True: one optional local call in L11
ROOT = Path.cwd()
assert (ROOT / 'data/lindistflow3_network.json').exists(), 'Start
   Jupyter in the tutorial folder'
```

- Leave both switches False. Python calculates new results; Llama replies come from saved records.
- If data are missing, rerun setup.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

L1: does the result match the request?

Classroom Labs / L1 / classroom-004

Executed notebook output

L1 · Does the evidence match?

classroom-004 · output 0 (zero-based) · recorded execution 3

	test	binding_pass
0	same request + same command	True
1	changed load; old evidence	False
2	changed command; old evidence	False

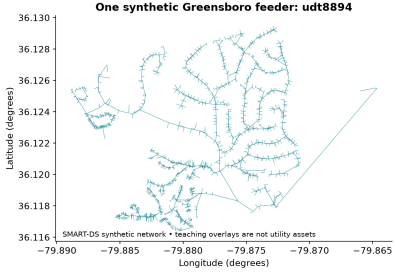
Reference L1 output. Only the unchanged request and command match.

- Read binding_pass. Expect True, False, False.
- Changed inputs or commands need a new check.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

Case 1: voltage support

Classroom Labs / L2 / classroom-007

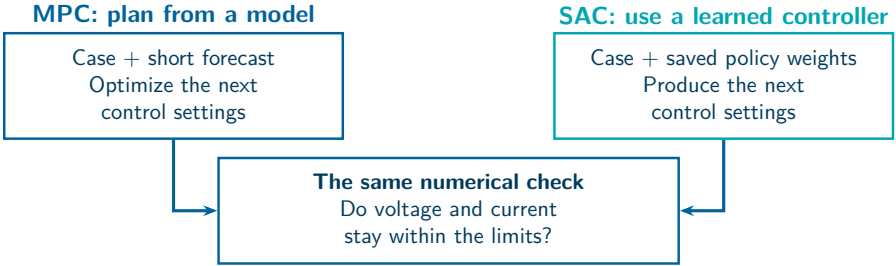


Synthetic SMART-DS feeder used for teaching.

- MPC plans ahead. SAC uses a saved learned policy.
- Check limits first. Among passing options, a smaller score J is better.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Two ways to propose settings



These are two control tools. The outer LLM agent can choose which tool to request next.

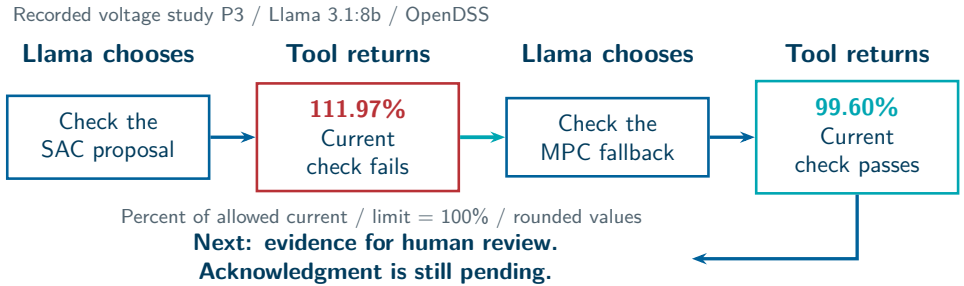
MPC = model predictive control. SAC = Soft Actor-Critic.

- MPC plans with a model. SAC proposes from learned weights. Both need the same electrical checks.

Source: Classroom Labs notebook. Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

A tool result changes the next step

Recorded P3 example / separate from the L2 calculation



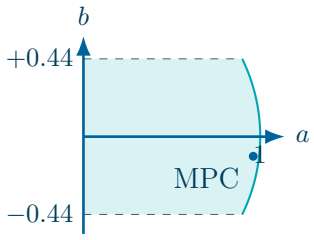
Saved P3 OpenDSS example, separate from L2. Values rounded.

- The first trial fails. Llama requests the fallback. A passing result goes to human review.

Source: Workshop recorded model replies and numerical study records. Voltage P3_current_reserve: Llama 3.1:8b planner, recovery and review calls 1–3. Public teaching archive. 27 September 2026.

Two control settings

Classroom Labs / L2 / classroom-007



$$p_i = aS_i, \qquad q_i = bS_i$$

$$0 \le a \le 1, \quad |b| \le 0.44, \quad a^2 + b^2 \le 1$$

$b < 0$: reactive absorption

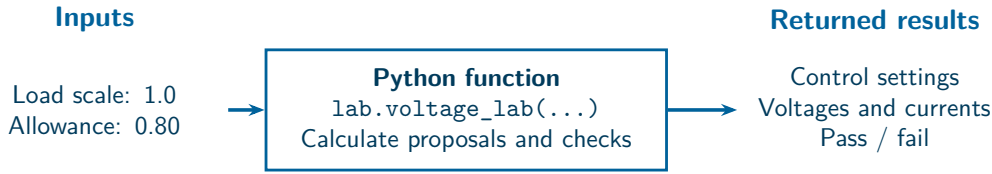
Inverter capability is one check.

Active and reactive power share the inverter capacity.

- a controls active power. b controls reactive power.
- Check inverter capacity, voltage and line current.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

A tool is a callable function



In L2, your cell calls this function directly.

In the recorded agent study, Python checks Llama's choice before running an allowed tool.

L2 calls lab.voltage_lab directly. Agent coordination is shown in the saved study.

- Inputs go in. Calculations run. Results come back. L2 calls this Python function directly.

Source: Classroom Labs notebook. Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

L2: run the default case

Classroom Labs / L2 / classroom-007

Classroom_Labs.ipynb L2 (classroom-007), source lines 1–4.

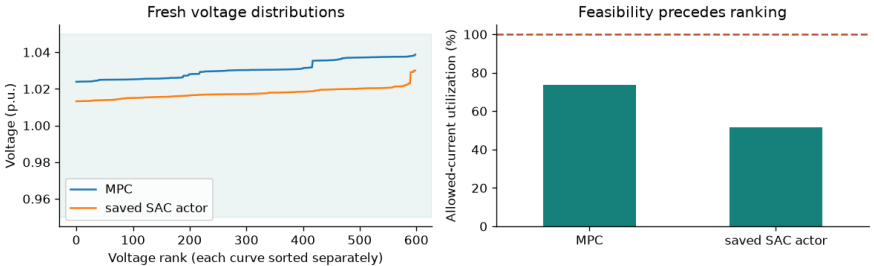
```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
started = time.perf_counter()
voltage = lab.voltage_lab(LOAD_SCALE, RATING_FRACTION)
```

- Run all of L2. It checks the MPC plan and the saved SAC policy.
- Read feasible, current_step_J and max_allowed_current_pct.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

L2: voltage and current plots

Classroom Labs / L2 / classroom-007



Reference L2 plots. Rerun for your inputs.

- Left: voltages should stay inside the allowed band.
- Right: current use should stay below 100% of the allowance.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

L2: both default proposals pass

Classroom Labs / L2 / classroom-007

Executed notebook output

L2 · Propose and check voltage settings

classroom-007 · output 1 (zero-based) · recorded execution 4

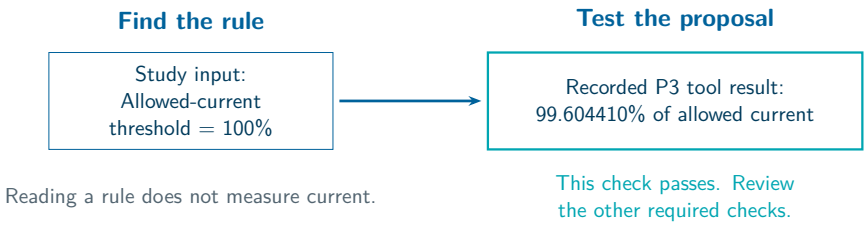
	method	feasible	current_ step_J	a	b	vmin	vmax	max_allowed_ current_pct	failures
0	MPC	True	0.031104	0.960307	-0.111114	1.023874	1.038709	73.619901	none
1	saved SAC actor	True	0.436428	0.340066	0.014024	1.013243	1.030000	51.679336	none

Reference L2: load scale 1.0, current allowance 0.80.

- MPC: 73.62% current use, score J about 0.03110.
- Saved SAC: 51.68% current use, score J about 0.43643.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

The limit and the calculated result



Example uses the separate saved P3 case.

- A rule states the limit. A tool checks the proposal against it.
- Keep both with the result so the operator can review it.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, pp.17–25; recorded P3 current check.

Exercise 1: lower the current allowance

Classroom Labs / L2 / classroom-007

```
RATING_FRACTION = 0.75
```

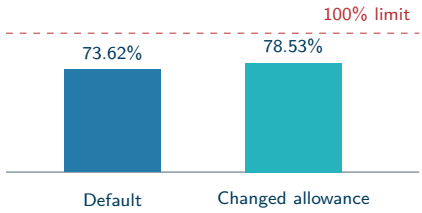
- In L2, change RATING_FRACTION from 0.80 to 0.75. Predict, then rerun.
- Compare feasible, current use and score J for both methods.
- Does this check cover the whole MV/LV network?

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Exercise 1: both still pass

L2: change one number

RATING_FRACTION
0.80 becomes 0.75
Predict, then rerun L2



Reference MPC results. Both checks still pass.

Reference L2 experiment. Both proposals pass at 0.75.

- A smaller allowance makes the same current a larger percentage.
- At 0.75: MPC uses 78.53%, SAC 54.85%. Both pass this test.

Source: Classroom Labs L2, cell classroom-007; verified default and exercise outputs.

L2: reset before continuing

Classroom Labs / L2 / classroom-007

Classroom_Labs.ipynb L2 (classroom-007), source lines 1–2.

```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
```

- Restore LOAD_SCALE=1.0 and RATING_FRACTION=0.80. Rerun L2.
- L3-L5 use a separate saved case called test_01.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

New calculations and saved replies



New calculation, saved reply and proposed pilot are different evidence types.

- Python calculates new results. A saved Llama reply keeps its original inputs.
- Your changed lab input does not change the saved reply.

Source: Classroom Labs notebook. Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

L3: inspect saved model choices

Classroom Labs / L3 / classroom-010

Classroom_Labs.ipynb L3 (classroom-010), source lines 1–2.

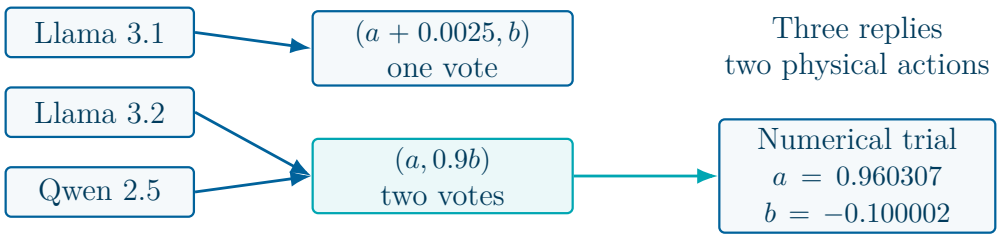
```
pool = lab.saved_pool('test_01')
display(pd.DataFrame(pool['responses'])[['model_id', 'operation_id
    ', 'reason', 'seconds', 'source']])
```

- Run L3. Read the case ID, chosen operations and recorded times.
- This cell displays old replies. It makes no new model call.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

Different words can request the same trial

Classroom Labs / L3 / classroom-010



Three saved replies request two different trials.

- The notebook groups identical tool requests and settings.
- A vote chooses what to test next. It does not prove that limits pass.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

L4: check the selected trial

Classroom Labs / L4 / classroom-013

Classroom_Labs.ipynb L4 (classroom-013), source lines 5–9.

```
started = time.perf_counter()
selected_trials = lab.trial_saved_selection(pool)
timings['L4 new verification of archived choices'] = time.
   perf_counter() - started
display(selected_trials['table'])
assert pool['frozen'] and pool['training_updates'] == 24
```

- Run L4 after L3. Python checks the selected trial for its original case.
- Compare majority voting with the saved LinUCB selector, which learns which trial to test.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

L4: read the new check

Classroom Labs / L4 / classroom-013

Executed notebook output

L4 · Check the selected trial

classroom-013 · output 1 (zero-based) · recorded execution 6

	selector	operation	new_check_passes	current_step_J	verification_s	failures
0	majority vote	RELAX_Q_10_PERCENT	True	0.031082	0.081254	none
1	frozen LinUCB	RELAX_Q_10_PERCENT	True	0.031082	0.111011	none

Saved choices, new checks of the original test_01 case.

- Both selectors choose the same trial and pass the new MV check.
- new_check_passes is the electrical check. Predicted reward is a learned score.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

L5: check model coverage

Classroom Labs / L5 / classroom-016

Classroom_Labs.ipynb L5 (classroom-016), source lines 1–4.

```
started = time.perf_counter()
scope_audit = lab.backend_audit(pool, run_opendss=RUN_OPENDSS)
timings['L5 selected model scope checks'] = time.perf_counter() -
    started
display(scope_audit['table'])
```

- Leave RUN_OPENDSS=False. Run L5 and read the table.
- The default checks the test_01 MPC plan with the MV model.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

A PASS has a scope

Classroom Labs / L5 / classroom-016

Executed notebook output

L5 · What does the model cover?

classroom-016 · output 0 (zero-based) · recorded execution 7

	model	command	fresh	passes	vmin	vmax	current_pct	failures
0	lindistflow3	same archived MV-MPC first action	True	True	1.023874	1.038709	73.661582	none

Default L5 checks MV only. OpenDSS is off.

- This result covers medium-voltage (MV) phases and currents.
- A full medium- and low-voltage (MV/LV) request needs the wider check or a HOLD.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

Saved voltage case: check SAC first

Recorded agent example / separate P3 study

RECORDED REPLAY · llama3.1:8b · voltage · call 1	
Input facts	Forecast load multipliers: [1.0, 0.94, 1.06]. Current allowance: 80% of nominal; voltage limits: [0.95, 1.05] p.u.
Llama choice	`VERIFY_SAC_FIRST`
Exact public reason	“A numerical SAC proposal is available; its fresh first-action check has not yet been requested.”
Recorded result	OpenDSS: 111.966607% of the allowed current; 2 current violations. Deterministic check: REJECT; failure = thermal.

Saved P3 case, separate from L2. Replay in L10B.

- Llama asks to check SAC. The tool finds a current-limit failure.

Source: Classroom Labs L10B: voltage replay, recorded call 1; full record in source guide.

Saved voltage case: try the MPC fallback

Recorded agent example / separate P3 study

RECORDED REPLAY · llama3.1:8b · voltage · call 2	
Input facts	The supplied failed checks are: thermal. An MPC plan already exists; its recovery-eligibility failure list is empty.
Llama choice	`USE_MPC_PLAN`
Exact public reason	“The MPC plan has been generated and is eligible for a numerical check. This is the next allowed step in the recovery process.”
Recorded result	OpenDSS: 99.604410% of allowed current; no recorded violations. Proposed P = 2387.157 kW; Q = +325.738 kvar.

Saved P3 case, separate from L2. Replay in L10B.

- Llama requests the existing MPC plan. The electrical tool checks it.

Source: Classroom Labs L10B: voltage replay, recorded call 2; full record in source guide.

Saved voltage case: return for review

Recorded agent example / separate P3 study

RECORDED REPLAY · llama3.1:8b · voltage · call 3

Input facts	The full-feeder first-action check passed; voltage is 1.001783–1.035026 p.u. Protection, transients and implementation authority remain outside this model.
Llama choice	`REVIEW_EVIDENCE`
Exact public reason	“The model has converged and the numerical evidence indicates that the control actions are within the allowed limits. Reviewing the evidence will provide more insight into the model's performance and the selected control actions.”
Recorded result	Numerical status: MODEL_RECOMMEND; recorded human acknowledgment: PENDING. Acknowledged = False; dispatch authorized = False; equipment operated = False.

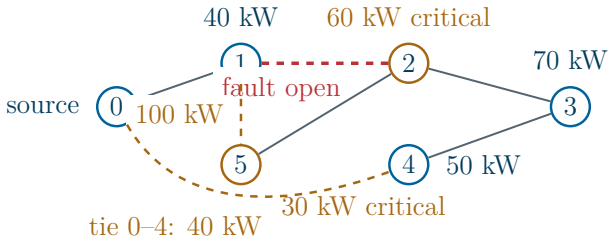
Saved P3 case, separate from L2. Replay in L10B.

- The checks pass. Human review is still pending. A fixed script gets the same final result.

Source: Classroom Labs L10B: voltage replay, recorded call 3; full record in source guide.

Case 2: restoring service

Classroom Labs / L6 / classroom-018



Critical demand
60 + 30 = 90 kW

Dashed ties:
options
not commands

Six-node active-power model. One branch is unavailable.

- Use the six-node teaching network with one faulted branch.
- Compare total service with service to critical loads.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Two service goals

Classroom Labs / L6 / classroom-018



Totals include node 1's 40 kW load upstream of tie 1–5.

Different goals can select different loads.

- One goal gives extra weight to critical loads.
- The other restores the most total kW. Both use the same network rules.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L6: solve the restoration problem

Classroom Labs / L6 / classroom-018

Classroom_Labs.ipynb L6 (classroom-018), source lines 1–5.

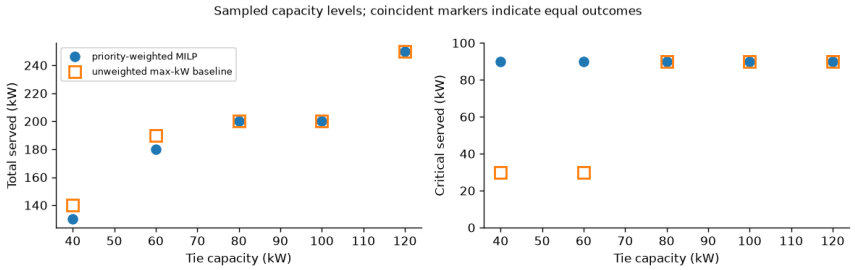
```
CAPACITIES_KW = (40., 60., 80., 100., 120.)
started = time.perf_counter()
restoration = lab.restoration_sweep(CAPACITIES_KW)
timings['L6 fresh MILP sweep + independent enumeration'] = time.
   perf_counter() - started
display(restoration['table'].drop(columns='pickup'))
```

- Run L6. MILP selects whole loads. A separate search checks its priority score.
- Read served_kw, critical_kw and MILP_matches_exhaustive_check.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L6: read both service plots

Classroom Labs / L6 / classroom-018



Reference L6 plots. Overlapping markers show equal outcomes.

- Left: total kW served. Right: critical kW served.
- Overlapping markers mean the two goals give the same result.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L6: the tradeoff at 40 kW

Classroom Labs / L6 / classroom-018

Executed notebook output

L6 · Which loads should return first?

classroom-018 · output 0 (zero-based) · recorded execution 8

Actual output header and rows 0–1 · tie_capacity_kw = 40.0

	tie_capacity_kw	method	served_kw	critical_kw	weighted_service	MILP_matches_exhaustive_check
0	40.0	priority-weighted MILP	130.0	90.0	940.0	True
1	40.0	unweighted max-kW baseline	140.0	30.0	410.0	True

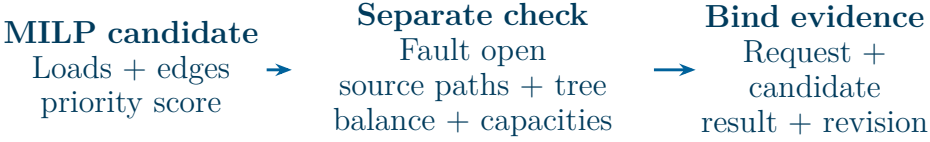
Reference L6: 40 kW varied tie, 100 kW fixed tie.

- Priority goal: 130 kW total, including 90 kW critical.
- Most-total-kW goal: 140 kW total, including 30 kW critical.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Restoration checks

Classroom Labs / L6 / classroom-018



Checks cover this graph model, not AC operation or field switching.

- Check source paths, branch capacity, power balance and the fault.
- A separate search checks the priority score.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Exercise 2: increase tie capacity

Classroom Labs / L6 / classroom-018

```
CAPACITIES_KW = (40., 60., 80., 100., 120., 180.)
```

- Add 180. to CAPACITIES_KW. Keep 40. first. Predict, then rerun L6.
- Compare total and critical kW at 40 and 180 kW.
- At which capacity do the two goals give the same result?

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Exercise 2: enough capacity removes this tradeoff

Classroom Labs / L6 / classroom-018

250 kW total / 90 kW critical

- At 180 kW, both goals serve 250 kW total and 90 kW critical.
- The priority goal helps at some limits. It does not improve every case.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L7: test an old result after a change

Classroom Labs / L7 / classroom-021

Classroom_Labs.ipynb L7 (classroom-021), source lines 1–3.

```
rest_record = restoration['records'][0]['weighted']
rest_request = {'task': 'restoration', 'capacity_kw':
   CAPACITIES_KW[0],
                'critical_nodes': [2, 5], 'model': '
   six_node_active_transport', 'revision': 1}
```

- Restore the original capacity list and rerun L6.
- Run L7. It changes the first request from 40 kW to 20 kW.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

L7: the old evidence no longer matches

Classroom Labs / L7 / classroom-021

Executed notebook output

L7 · Changed capacity needs a new check

classroom-021 · output 0 (zero-based) · recorded execution 9

```
Original contract: True
Changed rating with old evidence: False
Required response: HOLD old recommendation; request a fresh constrained solve.
```

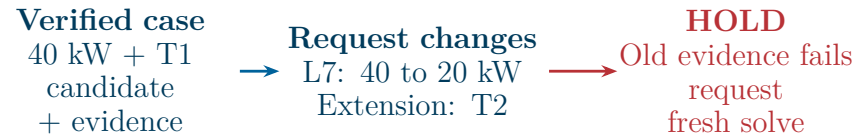
Reference L7. New inputs require new evidence.

- Original request: True. Changed request: False.
- HOLD the old recommendation and request a new solve.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

New inputs need new evidence

Classroom Labs / L7 / classroom-021



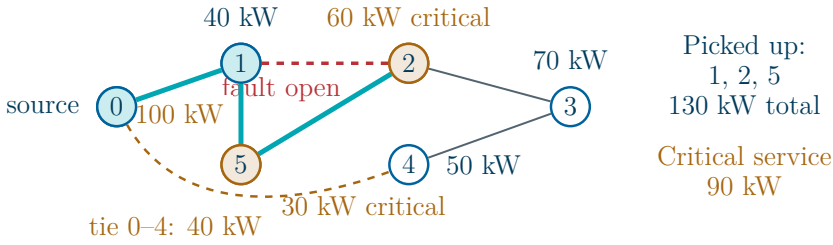
L7 changes capacity and revision, then tests the old evidence.

- An outdated check does not tell us whether the new plan works.
- Solve again, check again and attach the new result to the new request.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

The agent keeps the service goal

Classroom Labs / L6 / classroom-018



Proposed agent use. L6 itself runs Python.

- L6 runs Python directly. The operator sets priorities; MILP selects loads.
- An agent could request the solve and explain its returned tradeoff.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Restoration model limits

Classroom Labs / L6 / classroom-018

- This graph checks active power, not AC voltage or protection.
- Selected branches do not specify the order of field switching.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Saved restoration case: try another plan

Recorded agent example / separate feeder study

RECORDED REPLAY · llama3.1:8b · restoration · call 14

Input facts	E01 failed its thermal check; E02 passed. E02 serves 1242.777 kW with 65.938897% tie loading.
Llama choice	`SHED_075`
Exact public reason	“The candidate E01 has a thermal failure, and the lowest cost passed candidate E02 has a cost of -5509.135240000002. Selecting SHED_075 will test 75% disconnection of the ordered noncritical load objects, which may improve the cost.”
Recorded result	SHED_075 produces E03: 1843.614 kW served and 474.040 critical kW. E03 passes at 91.867215% tie loading; cost = -6109.971300 (smaller is better).

Saved feeder-catalog study, separate from L6. Replay in L10B.

- Llama requests a trial that restores more load and passes the current check.

Source: Classroom Labs L10B: restoration replay, recorded call 14; full record in source guide.

Saved restoration case: reject a failed trial

Recorded agent example / separate feeder study

RECORDED REPLAY · llama3.1:8b · restoration · call 15

Input facts	E03 is the lowest-cost passed candidate; one refinement evaluation remains. SHED_065 means disconnecting 65% of ordered noncritical load objects, not 65% of kW.
Llama choice	`SHED_065`
Exact public reason	“The candidate E01 has a thermal failure, and the lowest cost passed candidate E03 has a cost of -6109.9713, which is lower than the objective's target. Selecting SHED_065 will test 65% disconnection of the ordered noncritical load objects, which may improve the cost.”
Recorded result	E04 reaches 111.009413% tie loading: REJECT, thermal failure. The rejected trial has no eligible score; its larger served load cannot make it acceptable.

Saved feeder-catalog study, separate from L6. Replay in L10B.

- The next trial exceeds the current limit. Keep the previous passing plan.

Source: Classroom Labs L10B: restoration replay, recorded call 15; full record in source guide.

Saved restoration case: correct the ranking

Recorded agent example / separate feeder study

RECORDED REPLAY · llama3.1:8b · restoration · call 16

Input facts	E02 cost = -5509.135240; E03 cost = -6109.971300; smaller is better. Only the passed candidates E03 and E02 are offered for ranking.
Llama choice	`E02, E03`
Exact public reason	“The candidates were ranked based on their scores, with the lowest score (best) first. E02 has a score of -5509.135240000002, and E03 has a score of -6109.971300000003.”
Recorded result	Returned language order: E02 → E03. Correct numerical order: E03 → E02. The guard retains E03 and rechecks it as E05: 1843.614 kW served; no recorded failures.

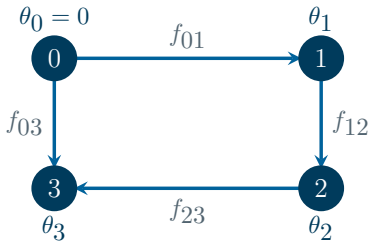
Saved feeder-catalog study, separate from L6. Replay in L10B.

- Python corrects the ranking and rechecks the lowest-cost passing plan.

Source: Classroom Labs L10B: restoration replay, recorded call 16; full record in source guide.

Case 3: suspicious readings

Classroom Labs / L8 / classroom-023



$x = (\theta_1, \theta_2, \theta_3)^\mathsf{T}$

4 flows + 3 injections + 1 angle

8 readings; $\mathrm{rank}(H) = 3$

Residual degrees: $8 - 3 = 5$

Fixed linear DC model

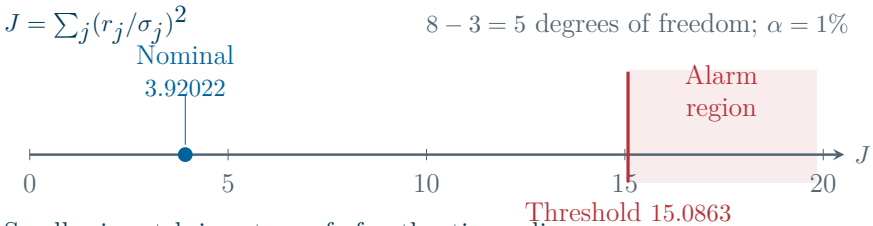
Four-bus DC model. Simulation truth is used only for offline comparison.

- Use the separate four-bus DC teaching model.
- Compare the fit to readings with error against known simulation truth.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

What the residual measures

Classroom Labs / L8 / classroom-023



Small mismatch is not proof of authentic readings

A good fit does not authenticate the readings.

- Weighted least squares (WLS) fits the readings. Residual J measures the mismatch.
- The true state is known in this simulation, not in a live control room.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: run two data attacks

Classroom Labs / L8 / classroom-023

Classroom_Labs.ipynb L8 (classroom-023), source lines 1–5.

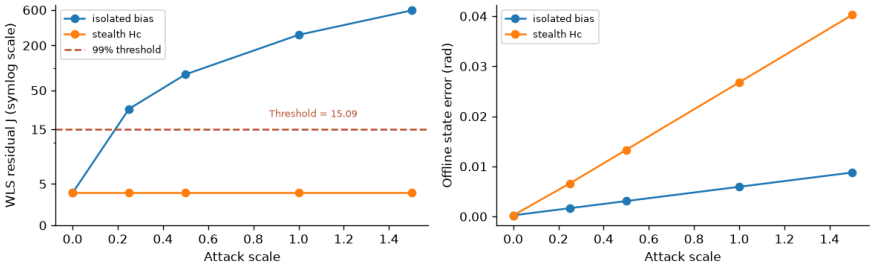
```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
timings['L8 fresh WLS attack sweep'] = time.perf_counter() -
   started
display(cyber['table'])
```

- Run L8 with the default scales.
- Compare one biased reading with a coordinated change across readings.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: read both cyber plots

Classroom Labs / L8 / classroom-023



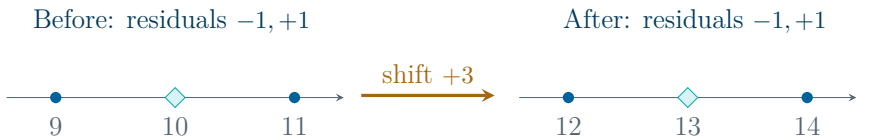
Reference L8 plots. State error uses known simulation truth.

- Left: mismatch J and the alarm threshold. Right: error against simulation truth.
- A quiet alarm can appear alongside a growing state error.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: a coordinated attack escapes the alarm

Classroom Labs / L8 / classroom-023



Values from the eight-reading L8 model.

- At scale 1, one biased reading gives J=277.50 and an alarm.
- The coordinated attack gives J=3.92, no alarm, and 0.02678 rad of state error.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Exercise 3: test a small bias

Classroom Labs / L8 / classroom-023

```
ATTACK_SCALES = (0., .10, .25, .5, 1., 1.5)
```

- Add .10 to ATTACK_SCALES. Predict, then rerun L8.
- At scale 0.10, compare J, the alarm flag and state error.
- Does no alarm mean the data are correct?

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Classroom Labs / L8 / classroom-023

$10.14 < 15.09$ to no residual alarm

- At scale 0.10, J is about 10.14, below the 15.09 alarm threshold.
- A quiet alarm alone cannot prove that readings are true.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: reset before recovery

Classroom Labs / L8 / classroom-023

Classroom_Labs.ipynb L8 (classroom-023), source lines 1–3.

```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
```

- Restore the original attack scales and rerun L8.
- L9 uses a separate fixed recovery case.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L9: one fixed recovery case

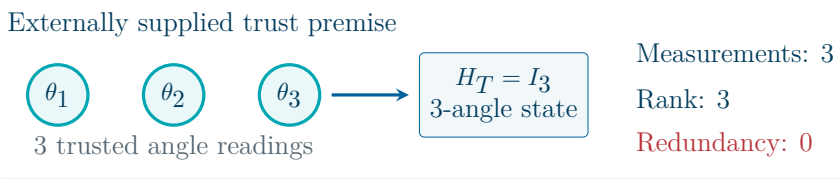
Classroom Labs / L9 / classroom-026

- Compare the same case with and without readings assumed to be trusted.
- Your L8 scale change does not change this case.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Data trust is an assumption

Classroom Labs / L9 / classroom-026



Three extra angle readings are assumed trustworthy.

- This example assumes three extra angle readings are trustworthy.
- Neither WLS nor Llama authenticates them.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

L9: HOLD or PROPOSE

Classroom Labs / L9 / classroom-026

Classroom_Labs.ipynb L9 (classroom-026), source lines 1–2.

```
without_trust = lab.trusted_response(cyber, protected_available=
    False)
with_trust = lab.trusted_response(cyber, protected_available=True)
```

- Run all of L9. Read status first.
- Then read the proposed injection and the offline violation count.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

L9: two outcomes

Classroom Labs / L9 / classroom-026

Executed notebook output

L9 · When can we propose a correction?

classroom-026 · output 0 (zero-based) · recorded execution 11

	condition	status	u_pu	offline_actual_violations
0	No trusted replacement channels	HOLD	NaN	NaN
1	Three trusted channels assumed	PROPOSE	0.121694	0.0

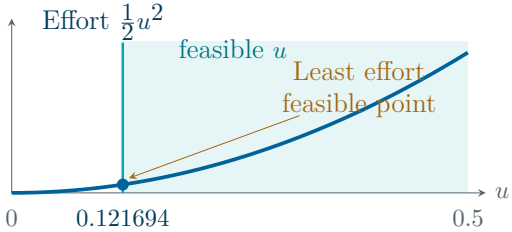
Reference L9 output for the fixed recovery case, not the edited L8 scale.

- Without trusted replacement readings: HOLD.
- With assumed trusted readings: PROPOSE, u about 0.121694 p.u., zero offline model violations.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

L9: check the correction

Classroom Labs / L9 / classroom-026



All four branches:

$|\hat{f}(u)| \leq \ell - 0.01$

$0 \leq u \leq 0.5$ p.u.

Fixed DC sensitivities

Fixed L9 correction: compare the predicted branch flow with the offline truth check.

- A quadratic-program (QP) solver finds the smallest allowed injection that meets the branch limits.
- One flow: predicted 0.17000 p.u., offline actual 0.17046 p.u., limit 0.18000 p.u.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

A proposal depends on its assumptions

Classroom Labs / L9 / classroom-026



A numerical proposal still needs human review.

- Assumed trusted data support the estimate. The solver calculates a correction.
- The operator still reviews data trust, model coverage and missing checks.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Saved cyber case: request the allowed tool

Recorded agent example / separate recovery study

RECORDED REPLAY · llama3.1:8b · cyber · call 7	
Input facts	The structured task already authorizes WLS_TRUSTED_REDISPATCH. Trusted sensors are explicitly assumed; no live utility measurements are supplied.
Llama choice	`WLS_TRUSTED_REDISPATCH`
Exact public reason	“This tool is authorized for the given use case and scope, and it is designed for a specific DC redispatch example with trusted measurements.”
Recorded result	The numerical tool proposes u = 0.121693667 p.u.; predicted flow12 = 0.170000 p.u. Offline synthetic-truth flow12 = 0.170461 p.u.; rating = 0.180000 p.u.

Saved four-bus study. Replay in L10B.

- Llama selects the supplied recovery tool. Python calculates the correction.

Source: Classroom Labs L10B: cyber replay, recorded call 7; full record in source guide.

Saved cyber case: keep human review

Recorded agent example / separate recovery study

RECORDED REPLAY · llama3.1:8b · cyber · call 8

Input facts	The model receives the actual numerical summary and passed_modeled_checks = True. No communication authentication, AC feeder or operating-permission check is included.
Llama choice	`REVIEW`
Exact public reason	“The provided JSON output indicates that the numerical workflow has completed successfully, and the results are within the expected scope. The 'passed_modeled_checks' field is set to True, indicating that the modeled checks have been passed. Therefore, the choice is to REVIEW the results to ensure they are accurate and within the expected range.”
Recorded result	Numerical verification: PASS. Language choice: REVIEW. Overall result: REVIEW. Human acknowledgment: PENDING; dispatch authorized = False; equipment operated = False.

Saved four-bus study. Replay in L10B.

- The model check passes. The operator must still review trust and missing checks.

Source: Classroom Labs L10B: cyber replay, recorded call 8; full record in source guide.

L10: inspect the five-role workflow

Classroom Labs / L10 / classroom-028

- Run L10 to see every recorded stage and result.
- The stages share case facts, proposals and check results. The next slides explain their jobs.

Source: Classroom_Labs.ipynb, L10–L11; cells classroom-028 and classroom-030.

L10B: replay the three Llama examples

Classroom Labs / L10B / classroom-llama-walkthrough

Classroom_Labs.ipynb L10B (classroom-llama-walkthrough), source lines 1–5.

```
SCENARIO = "voltage"  # Change to "restoration" or "cyber", then
   rerun this cell.
from IPython.display import HTML, display
from src.llama_walkthrough import load_episodes,
   render_episode_html
replay_episodes = load_episodes(ROOT)
display(HTML(render_episode_html(replay_episodes[SCENARIO])))
```

- Set SCENARIO to voltage, restoration and cyber. Rerun for each.
- These are saved replies and tool results. No new Llama call runs here.

Source: Classroom_Labs.ipynb, L10B; cell classroom-llama-walkthrough; src/llama_walkthrough.py.

L12: practice operator readback

Classroom Labs / L12 / classroom-033

Classroom_Labs.ipynb L12 (classroom-033), source lines 9–10.

```
wrong_id = lab.acknowledge(mv_scope, 'wrong-id', 'Student operator
    ')
readback = lab.acknowledge(mv_scope, mv_scope['recommendation_id
    '], 'Student operator')
```

- Leave L11 disabled, then run L12.
- Check the wrong ID, the matching ID and equipment_dispatch=False.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

L12: approval must match the checked scope

Classroom Labs / L12 / simulated readback

Executed notebook output

L12 · Match the operator readback

classroom-033 · output 0 (zero-based) · recorded execution 15

Contiguous columns: state through equipment_dispatch; original row order and row index.

	state	evidence_ matches	engineering_ pass	scope_ matches	scope	acknowled ged	equipment_ dispatch
0	HOLD	True	True	False	MV phases only	False	False
1	READY_FOR_ READBACK	True	True	True	MV phases only	True	False

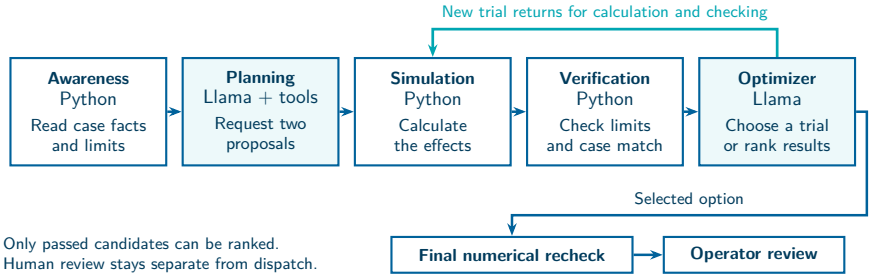
Reference L12 output. Readback is simulated.

- Full MV/LV request: HOLD. The MV teaching request allows a simulated readback.
- A wrong ID fails. This exercise never sends equipment commands.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Our multi-agent design: five roles

Recorded five-role study / L10



Default Colab: replay of saved roles and results. No new LLM run.

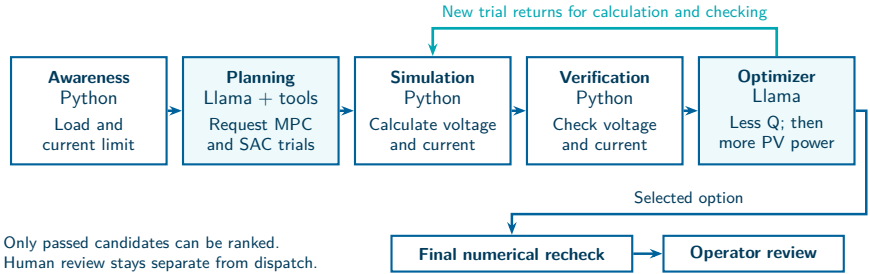
Custom Python workflow, one Llama model in two roles. L10 displays its saved trace.

- The roles pass structured results to each other. Llama handles planning and refinement; Python runs the other stages.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Five roles in the voltage case

Recorded five-role study / L10



Final recheck: 50.75% allowed-current use. MV only. Human review pending.

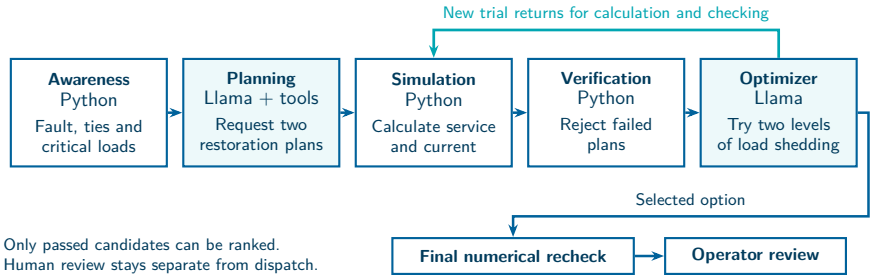
L10 MV voltage archive, separate from L2 and the P3 replay in L10B.

- Llama asks for less reactive power, then more solar output. Python checks each new trial.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Five roles in the restoration case

Recorded five-role study / L10



Selected: 1,843.61 kW served, 474.04 kW critical, 91.87% current use.

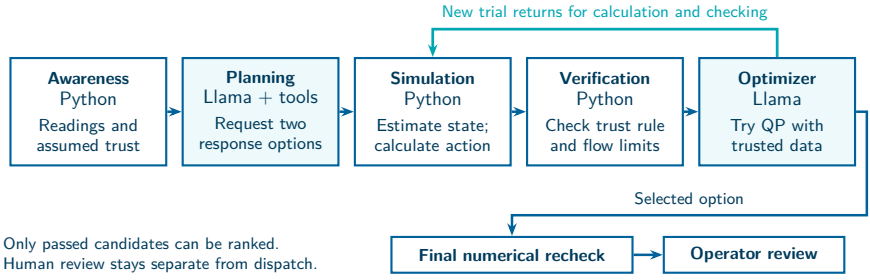
L10 feeder catalog, separate from the six-node L6 exercise.

- The goal stays fixed. A failed trial cannot enter the final ranking, and Python corrects a wrong cost order.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Five roles in the cyber case

Recorded five-role study / L10



Selected correction: u = 0.121694 p.u. Predicted flow = 0.17000 p.u.

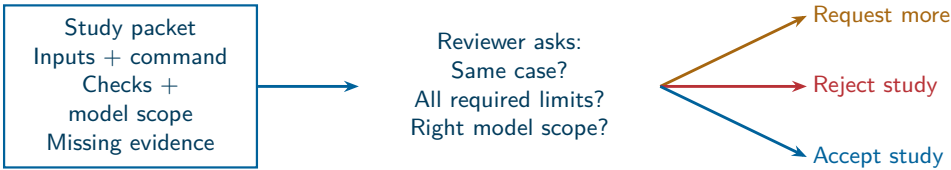
L10 four-bus DC cyber_stealth archive. Trusted channels are assumed.

- The workflow keeps the trust assumption visible. It cannot create trusted data from a convincing answer.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

The operator reviews one evidence packet

Proposed advisory pilot / teaching design



Accepted study ≠ permission to operate equipment.

Proposed advisory pilot — teaching design

Proposed control-room use. Study review does not operate equipment.

- Review the inputs, proposal, worst margins and missing checks.
- Request another study, reject the proposal or accept it for the next authorized step.

Source: Teaching proposal informed by NIST OT guidance and the cited notebook checks.

What the evidence shows

Proposed advisory pilot / teaching design



Compare with a fixed script + the same tools. The owner can stop the pilot.

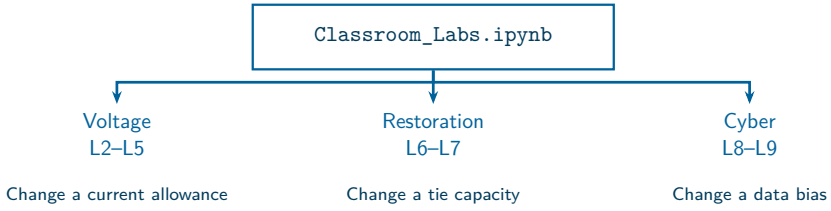
Proposed advisory pilot — teaching design

L10 comparison: same tools and trial budgets. No general performance gain is established.

- In L10, five paired final costs tied; one Llama route was worse.
- Start with past cases and read-only use. Measure benefit against the same tools in a fixed script.

Source: Teaching proposal informed by NIST OT guidance and the cited notebook checks.

Questions and your next lab run



L10B replays Llama choices in the same file. Each case uses a separate teaching network.

Keep the saved replies separate from your new experiment.

Open the three cases in Colab

- Choose one case. Change one input, rerun and save the result.
- Explain the agent choice, tool result, check and operator decision.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.